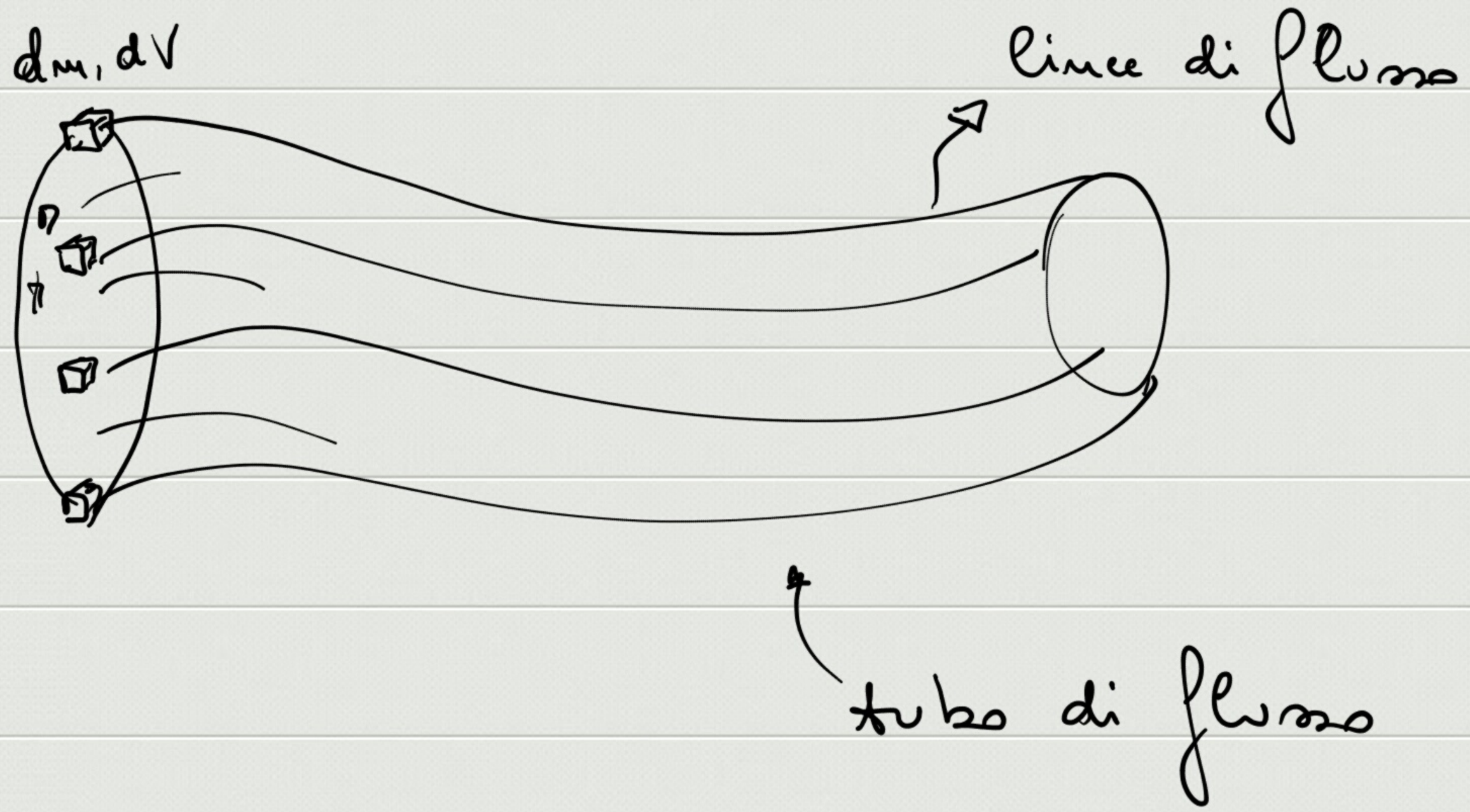
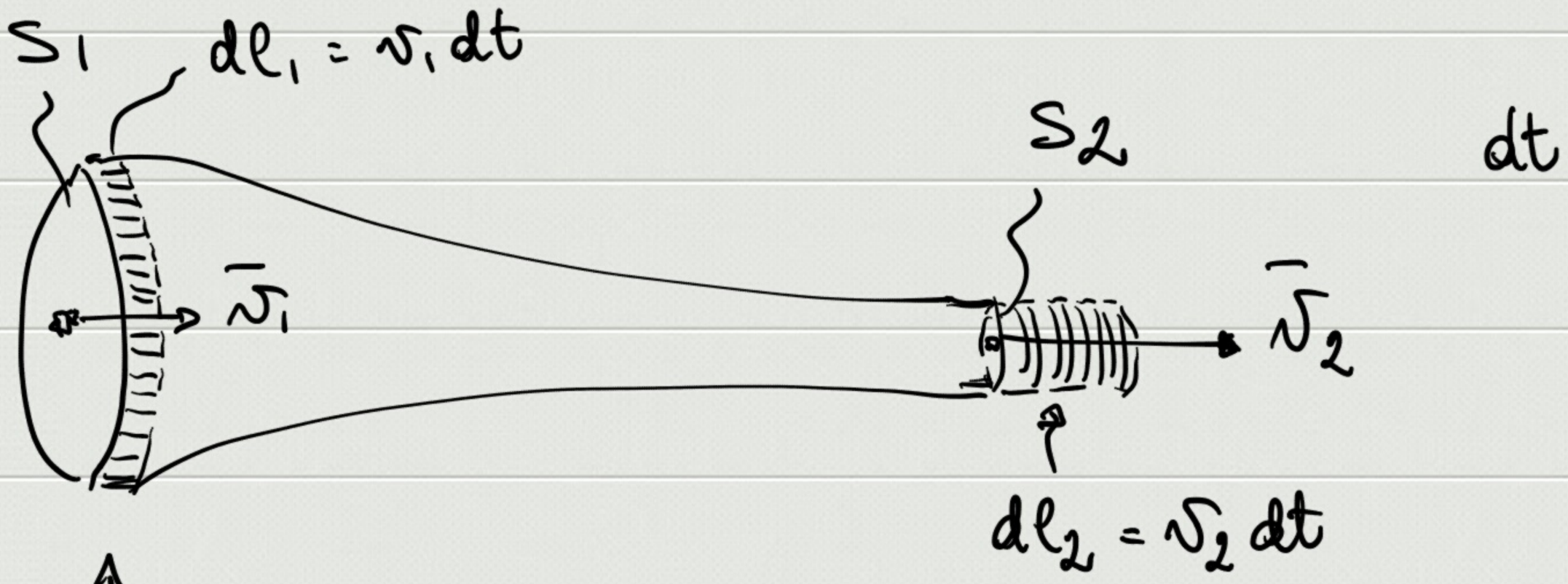
Fluido ideale : $\eta = 0$ $\rho = cost$

Moto laminare $\Rightarrow$ no moti "vorticosi"

Moto "stazionario"





$$dm_1 = \rho\, dV_1 = \rho S_1 v_1 dt$$

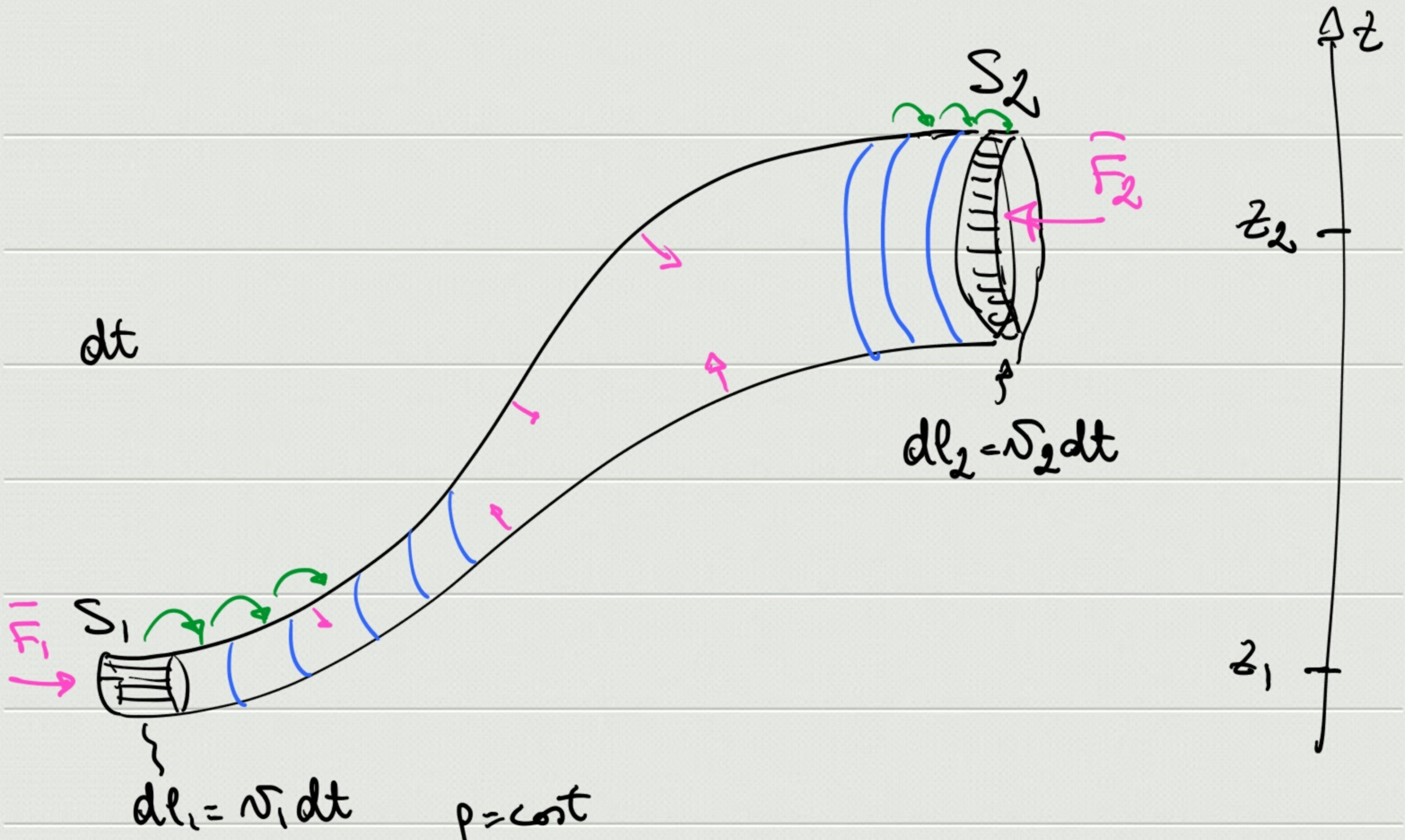
$$dm_2 = \rho\, dV_2 = \rho S_2 v_2 dt$$

$$dm_1 = dm_2 \Rightarrow \boxed{v_1 S_1 = v_2 S_2}$$

portata $\boxed{q = vS = \text{costante}}$

$$\boxed{v_m = \frac{1}{S}\int_S v\, dS}$$

$$dq = v\, dS \Rightarrow q = \int_S dq = \int_S v\, dS = v_m S$$



$\rho = cost$

$$dm_1 = dm_2 \Rightarrow dV_1 = dV_2 = dV$$

$$\boxed{dE_k = dW_{TOT}}$$

$$dW_{TOT} = \underset{\text{peso}}{dW_{vol}} + \underset{\text{pressione}}{dW_{sup}}$$

$$dW_{vol} = dW_{peso} = -dE_{p,peso} \qquad *$$

$$-(\cancel{E'_{p_1}} - E_{p_1}) - (\cancel{E'_{p_2}} - \cancel{E_{p_2}}) - (E'_{p,3} - \cancel{E_{p,3}}) - \dots$$

$$* = -(dm\,g\,z_2 - dm\,g\,z_1) = -\rho\,dV\,(z_2 - z_1)$$

$$dW_{sup} = \bar{F}_1 \, d\bar{l}_1 + \bar{F}_2 \, d\bar{l}_2 = p_1 S_1 \, dl_1 - p_2 S_2 \, dl_2 =$$

$$= (p_1 - p_2) \, dV$$

$$dm = \rho \, dV$$

$$dE_k = \frac{1}{2} dm \, v_2^2 - \frac{1}{2} dm \, v_1^2 = \frac{1}{2} \rho \, dV (v_2^2 - v_1^2)$$

$$\frac{1}{2} \rho (v_2^2 - v_1^2) = -\rho (z_2 - z_1) + (p_1 - p_2)$$

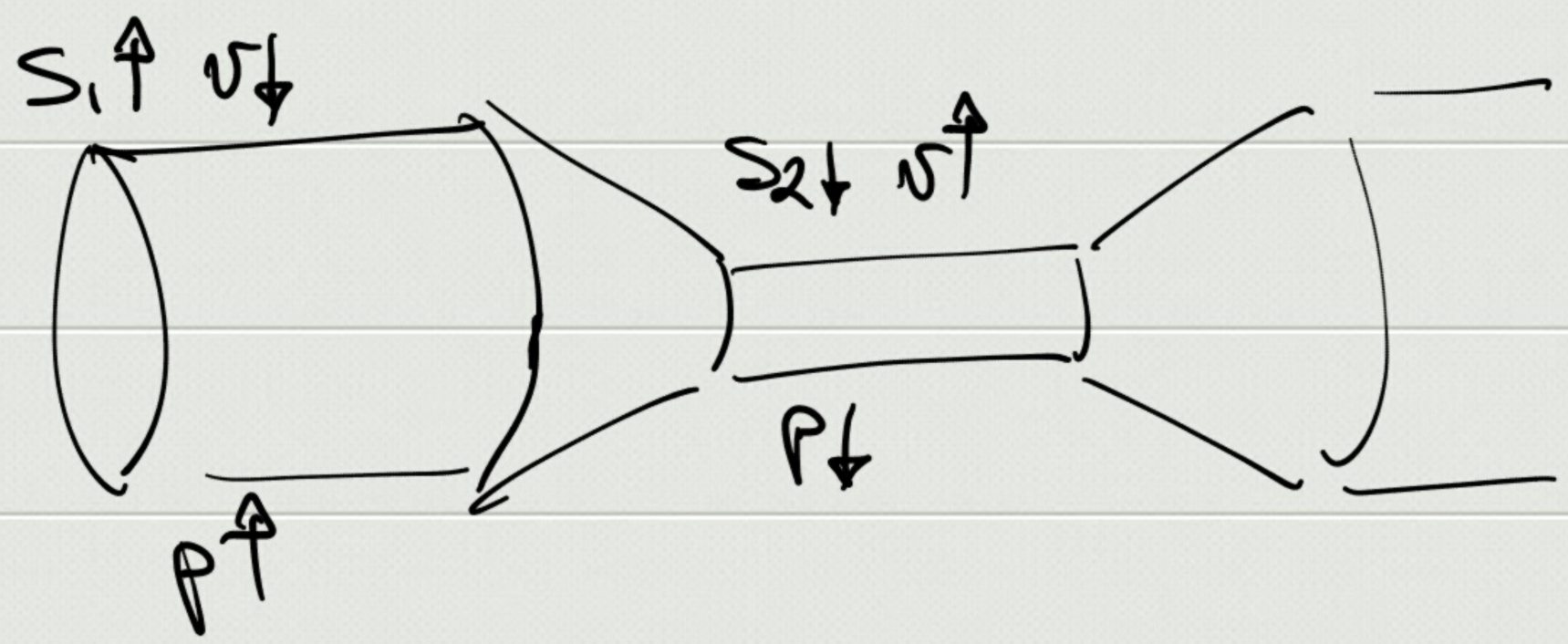
$$p_1 + \rho z_1 + \frac{1}{2} \rho v_1^2 = p_2 + \rho z_2 + \frac{1}{2} \rho v_2^2$$

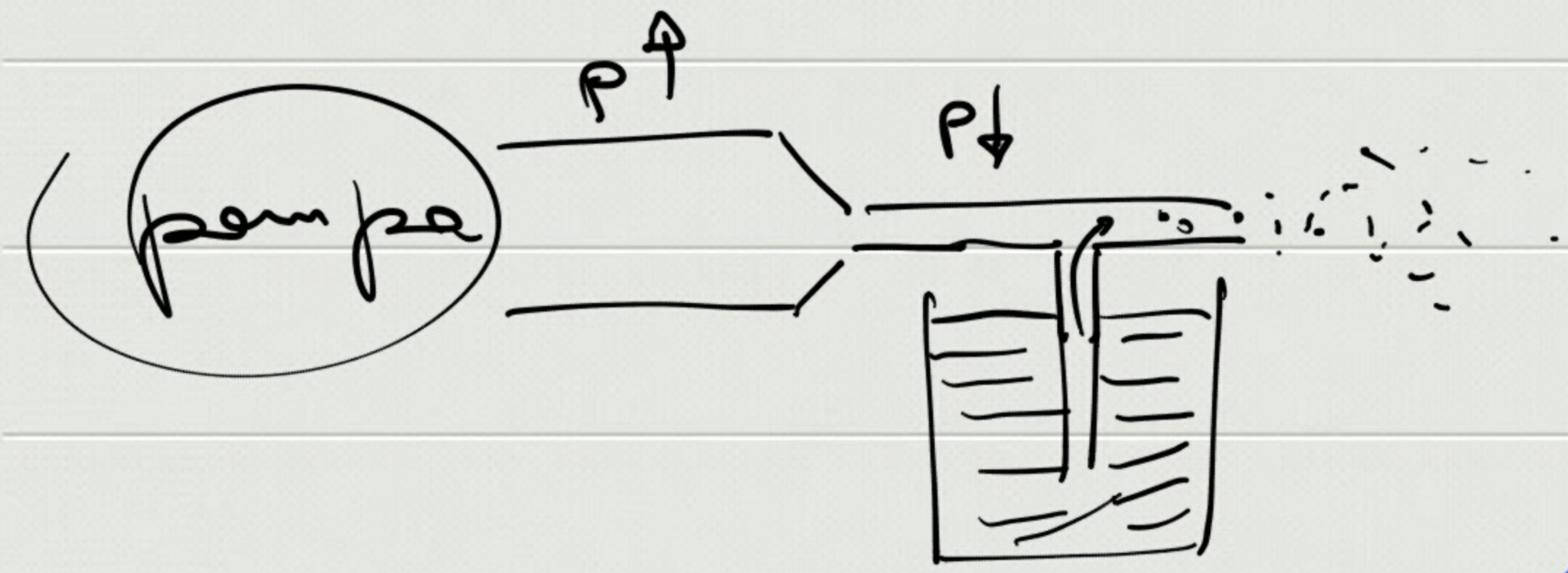
$$\Rightarrow \boxed{p + \rho z + \frac{1}{2} \rho v^2 = \text{cost}}$$

Teorema di Bernoulli

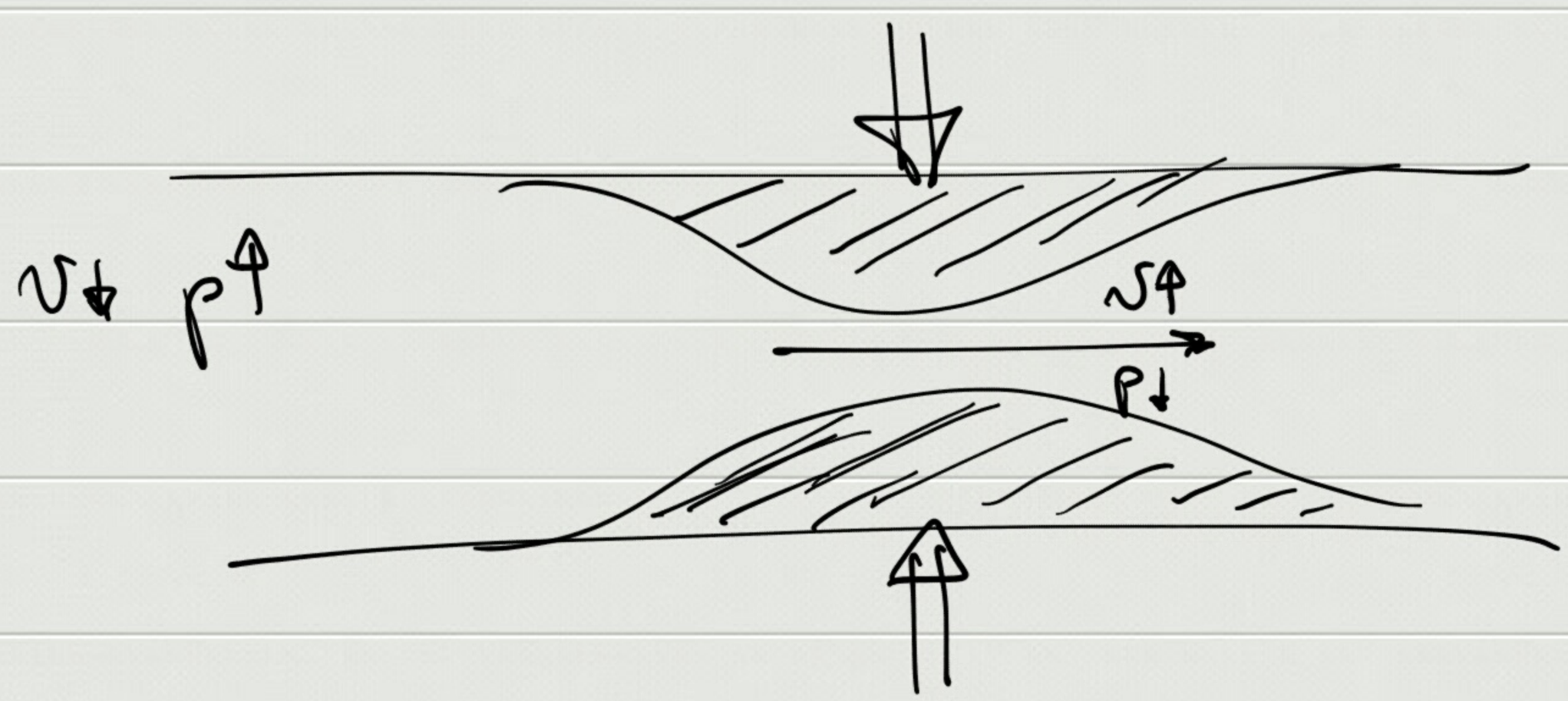
$z = \text{cost}$

$q = vS = \text{cost}$




pompa
P↑
P↓

P↓ v↑
P↑ v↓


v↓ P↑
v↑
P↓